The classification trichotomy for chiral algebras: three invariants and four shadow classes

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Abstract

Three numerical invariants of a chiral algebra $\mathcal{A}$ (the generator OPE pole order $p_{\max}(\mathcal{A})$, the collision depth $k_{\max}(\mathcal{A})$, and the shadow depth $r_{\max}(\mathcal{A})$) measure three structurally distinct features and need not coincide. The relation $k_{\max} = p_{\max} - 1$ always holds, a consequence of the $d\log$ absorption of the bar propagator, but $r_{\max}$ is independent of $p_{\max}$. The $\beta\gamma$ system is the archetypal witness: $p_{\max} = 1$, $k_{\max} = 0$, $r_{\max} = 4$. This distinction yields a trichotomy at degree 2 (trivial / multiplication / differential) governing the operator order of commuting Hamiltonians, and a four-class partition G, L, C, M of the standard landscape by shadow depth. All claims are verified computationally across 14 engine modules and 929 tests.

1 Introduction

Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ carries a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra, and the projections of $\Theta_{\mathcal{A}}$ to finite degree form the shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$. Three invariants extracted from this data measure three different structural features of $\mathcal{A}$.

The *generator OPE pole order* $p_{\max}(\mathcal{A})$ records the maximal singular order of the OPE among generators. The *collision depth* $k_{\max}(\mathcal{A})$ records the maximal pole order of the degree-2 collision residue $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$. The *shadow depth* $r_{\max}(\mathcal{A})$ records the degree at which the tower terminates (or ∞ if it does not).

These invariants are distinct projections of common ordered data. The collision kernel $K^{\mathrm{coll}}_{\mathcal{A}}(z)$ lives on the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, the cofree tensor coalgebra with deconcatenation coproduct; it is an element of the E_1 convolution algebra that carries the full spectral-parameter dependence. Its homotopy coinvariant $q_2(K^{\mathrm{coll}}_{\mathcal{A}})$ retains an operator class, and the normalized functional gives $\tau^{\mathrm{res}}_{\mathcal{A},2} q_2(K^{\mathrm{coll}}_{\mathcal{A}}) = \kappa$ for abelian and Virasoro-type families. On the non-abelian affine row this trace is κ_{dp}, with Sugawara contribution $\dim(\mathfrak{g})/2$. This scalar trace, and the shadow depth $r_{\max}$ is computed from the Σ_n-coinvariant tower. Both are proper quotients of the ordered data.

The bar propagator $d\log(z - w) = dz/(z - w)$ absorbs one power of the OPE singularity, so $k_{\max} = p_{\max} - 1$ always. But the shadow depth $r_{\max}$ involves composite-field contributions at higher degrees and is not determined by the generator OPE poles. The $\beta\gamma$ system has $p_{\max} = 1$ (simple pole only) yet $r_{\max} = 4$ (the quartic contact class from the composite $:\beta\gamma:$ does not vanish). The three invariants disagree maximally.

This paper formalizes the distinction and extracts two classification results: the operator-order trichotomy at degree 2, and the four-class partition of the standard landscape.

2 The $\beta\gamma$ witness

The $\beta\gamma$ system is generated by fields $\beta(z), \gamma(z)$ of conformal weights 1 and 0 with OPE

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}, \qquad \gamma(z)\beta(w) \sim \frac{-1}{z-w}, \qquad \beta\beta \sim 0, \qquad \gamma\gamma \sim 0.$$

The maximal OPE pole among generators is a simple pole, so $p_{\max}(\beta\gamma) = 1$.

By the $d\log$ absorption principle, a simple OPE pole is absorbed entirely by the bar propagator kernel $d\log(z-w)$ and produces no collision residue. The collision depth is therefore $k_{\max}(\beta\gamma) = 0$. At degree 2, the $\beta\gamma$ system is invisible to the bar complex.

The composite field $:\beta\gamma:$ has conformal weight 1. Its self-OPE

$$(:\beta\gamma:)(z)\,(:\beta\gamma:)(w) \sim \frac{-1}{(z-w)^2}$$

has a second-order pole; this contributes at degree 4 to the shadow obstruction tower. The quartic contact class $\mathfrak{Q}_{\beta\gamma}$ is nonvanishing, so the shadow depth is $r_{\max}(\beta\gamma) = 4$. After degree 4, rank-one rigidity forces all higher-degree projections to vanish, and the tower terminates.

The three invariants for $\beta\gamma$ are

$$p_{\max} = 1, \qquad k_{\max} = 0, \qquad r_{\max} = 4.$$

This triple is the sharpest witness to the independence of the three invariants: the OPE pole order is minimal, the collision depth is trivial, yet the shadow depth is nontrivial.

3 Definitions

Definition 3.1 (Generator OPE pole order). Let $\mathcal{A}$ be a chiral algebra with minimal strong generators $\{a_\alpha\}$. The *generator OPE pole order* is

$$p_{\max}(\mathcal{A}) := \max_{\alpha,\beta} \left\{ n \geq 0 \;\middle|\; a_\alpha(z)a_\beta(w) \text{ has a nonzero } (z-w)^{-n} \text{ term} \right\}.$$

Definition 3.2 (Collision depth). The *collision depth* of $\mathcal{A}$ is

$$k_{\max}(\mathcal{A}) := \max\left\{ k \geq 0 \;\middle|\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \text{ has a nonzero } z^{-k} \text{ term} \right\}.$$

Definition 3.3 (Shadow depth). The *shadow depth* of $\mathcal{A}$ is

$$r_{\max}(\mathcal{A}) := \max\left\{ r \geq 2 \;\middle|\; \Theta^{\leq r}_{\mathcal{A}} \text{ has a nonzero } r\text{-degree projection} \right\} \in \{2, 3, 4, \infty\}.$$

For $r_{\max} < \infty$, the tower terminates at degree $r_{\max}$ and all higher projections vanish. For $r_{\max} = \infty$, the tower has nontrivial components at every degree.

Proposition 3.4 (Fundamental relation). *For every chiral algebra $\mathcal{A}$,*

$$k_{\max}(\mathcal{A}) = p_{\max}(\mathcal{A}) - 1.$$

Proof. The bar propagator extracts collision residues against $d\log(z-w) = dz/(z-w)$. A pole of order n in the OPE contributes a residue of order $n-1$ after the $d\log$ kernel absorbs one power of $(z-w)$. □

Proposition 3.5 (Independence of $r_{\max}$ from $p_{\max}$)**.** *The shadow depth $r_{\max}$ is not determined by the generator OPE pole order $p_{\max}$. The pair $(p_{\max}, r_{\max})$ attains the values $(1,4)$, $(2,2)$, $(2,3)$, $(4,\infty)$, and $(2N,\infty)$ for $N \geq 3$ in the standard landscape.*

Proof. The witnesses are: $\beta\gamma$ for $(1,4)$; Heisenberg for $(2,2)$; affine $\widehat{\mathfrak{g}}_k$ of type A_1 for $(2,3)$; Virasoro for $(4,\infty)$; and $\mathcal{W}_N$ for $(2N,\infty)$. In each case the shadow depth is computed from the explicit shadow obstruction tower. The pair $(2,2)$ vs. $(2,3)$ shows that $p_{\max} = 2$ does not determine $r_{\max}$; the pairs $(4,\infty)$ and $(1,4)$ show that the inequality can go in either direction. □

4 The operator-order trichotomy

The collision depth $k_{\max}$ classifies chiral algebras into three regimes through the operator order of the commuting Hamiltonians $H_i = \mathrm{Sh}_{0,n}(\Theta_{\mathcal{A}})_i$ on genus-0 n-point moduli.

Theorem 4.1 (Operator-order trichotomy)**.** *For every chiral algebra $\mathcal{A}$ in the standard landscape, $k_{\max}(\mathcal{A}) \in \{0,1\} \cup \{3,4,5,\ldots\}$. The value $k_{\max} = 2$ is excluded by locality and conformal dimension. The three regimes are:*

$k_{\max}$	*Regime*	*Hamiltonians H_i*	*Examples*
0	*trivial*	*no commuting structure at degree 2*	*$\beta\gamma$, free fermion*
1	*multiplication*	*scalar/matrix multiplication*	*Heisenberg, affine KM*
≥ 3	*differential*	*differential operators of order $k_{\max}-1$*	*Virasoro, $\mathcal{W}_N$*

Proof. The generator OPE pole order $p_{\max}$ takes values in $\{1,2\} \cup \{4,5,\ldots\}$ for bosonic chiral algebras in the standard landscape: the OPE of two weight-h currents has poles at orders $0,1,\ldots,2h$, and the maximal nonzero pole is $2h$ for bosonic self-OPEs. Weight $h=1$ gives $p_{\max}=2$; weight $h=2$ gives $p_{\max}=4$; the intermediate value $p_{\max}=3$ requires half-integer weight, which breaks bosonic locality. Since $k_{\max} = p_{\max}-1$, the gap at $p_{\max}=3$ becomes a gap at $k_{\max}=2$.

The identification of the three regimes follows from the pole structure of the collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$: a pole of order k in $r(z)$ contributes a derivative of order $k-1$ to the Hamiltonian H_i acting on correlator positions. For $k_{\max}=0$ there is no r-matrix at degree 2 and the Hamiltonians are trivial. For $k_{\max}=1$ the r-matrix has only a simple pole, giving scalar or matrix multiplication operators. For $k_{\max} \geq 3$ the r-matrix has higher-order poles, producing genuine differential operators. □

Remark 4.2 (Why $k_{\max}=2$ is absent). The exclusion of $k_{\max}=2$ is not a convention but a consequence of the conformal-weight spectrum. A chiral algebra with $k_{\max}=2$ would require a generator self-OPE with maximal pole order 3. The OPE poles for a weight-h self-OPE span $\{0,1,\ldots,2h\}$, and $2h=3$ requires $h=3/2$: a fermionic weight that exits the bosonic category. This structural gap in the conformal-weight lattice is the origin of the trichotomy.

5 Two classifications, one fingerprint

The title of this paper refers to the trichotomy at degree 2 (Theorem 4.1): the three operator-order regimes $k_{\max} \in \{0, 1, \geq 3\}$ classifying the GZ26 commuting-Hamiltonian structure. A distinct classification, the quadrichotomy G, L, C, M by shadow depth $r_{\max} \in \{2, 3, 4, \infty\}$, appears in §6 below. The two classifications are independent and must not be conflated.

The $k_{\max}$-trichotomy is a three-cell partition (trivial / multiplication / differential) controlled by slot (1) of the canonical fingerprint and operative at bar degree 2.

The $r_{\max}$-quadrichotomy is a four-cell partition (G, L, C, M) controlled by slot (2) of the canonical fingerprint and operative across all bar degrees.

Neither refines the other: the trichotomy groups $\beta\gamma$ ($k_{\max} = 0$) with the free fermion, whereas the quadrichotomy places $\beta\gamma$ in class C and the free fermion in class G; conversely, the quadrichotomy groups Vir_c with $\mathcal{W}_N$ as class M, whereas the trichotomy discriminates them by $k_{\max} = 3$ against $k_{\max} = 2N - 1$. Both classifications are projections of the canonical fingerprint

$$\varphi(\mathcal{A}) \;=\; (p_{\max},\, r_{\max},\, \chi_{\mathrm{VOA}},\, n_{\mathrm{strong}},\, \mathrm{coset}),$$

which is a complete invariant of the Koszul-bar-complex structure. Accordingly, the title *classification trichotomy* refers exclusively to Theorem 4.1 (three regimes at degree 2); the four shadow classes below are the companion quadrichotomy, and a fifth class FF at $\kappa = 0$ completes the fingerprint stratum.

6 The four shadow classes G, L, C, M

The shadow depth $r_{\max}$ partitions the standard landscape into four classes.

Definition 6.1 (Shadow-depth classification)**.** A chiral algebra $\mathcal{A}$ belongs to shadow class:

- G (Gaussian) if $r_{\max}(\mathcal{A}) = 2$;
- L (Lie/tree) if $r_{\max}(\mathcal{A}) = 3$;
- C (contact/quartic) if $r_{\max}(\mathcal{A}) = 4$;
- M (mixed/infinite) if $r_{\max}(\mathcal{A}) = \infty$.

Definition 6.2 (Class C: structural characterization)**.** A chirally Koszul algebra $\mathcal{A}$ is *contact (class C)* if its shadow tower has finite quartic depth:

$$S_3(\mathcal{A}) = 0, \qquad S_4(\mathcal{A}) \neq 0, \qquad S_r(\mathcal{A}) = 0 \quad (r \geq 5).$$

Equivalently, the averaged Maurer–Cartan projection has no cubic component, has a nonzero quartic contact component, and has no higher shadow components. The standard witnesses are bc_λ and $\beta\gamma_\lambda$, for which $S_4 = -5/12$.

Remark 6.3 (Why class C is not a pole-order condition). The tempting identification "class C = triple-pole $R(z)$" fails on the $\beta\gamma$ witness: $(p_{\max}, k_{\max}, r_{\max}) = (1, 0, 4)$ has a simple OPE pole, vanishing collision residue, and nonzero quartic shadow. Class C is not detected at bar degree 2; it is detected by the averaged quartic contact component.

Theorem 6.4 (Four-class stratification of the standard landscape)**.** *Let $\mathcal{A}$ be a logarithmic chirally Koszul algebra with conformal vector at non-critical level $\kappa(\mathcal{A}) \neq 0$. Then exactly one of the following holds:*

(i) $\mathcal{A}$ *is class G:* $r_{\max} = 2$*;* $S_r(\mathcal{A}) = 0$ *for* $r \geq 3$ *on every stratum; the shadow metric* $Q_L(t) = (2\kappa)^2$ *is a constant and* $\Delta = 0$ *trivially.*

(ii) $\mathcal{A}$ *is class L:* $r_{\max} = 3$*;* $S_3(\mathcal{A}) = \alpha \neq 0$ *on a primary line,* $S_4|_{\mathrm{prim}} = 0$ *(Jacobi identity), and the rank-one charged stratum is abelian (*$\mathfrak{Q} = 0$*); the tower terminates at degree 3 by the cubic-gauge-trivialisation argument.*

(iii) $\mathcal{A}$ *is class C: in the sense of Definition 6.2; the tower terminates at degree 4 because (C3) enforces charge conservation: the quartic pump* $[\mathfrak{Q}_{q_\star}, \mathfrak{Q}_{q_\star}]_{\mathrm{cyc}}$ *carries charge* $2q_\star$ *outside* $\mathrm{supp}(g^{\mathrm{mod},(0)})$*, hence vanishes identically in the cyclic complex.*

(iv) $\mathcal{A}$ *is class M:* $\alpha \neq 0$ *on a primary line and the single-line dichotomy with* $\Delta \neq 0$ *forces* $r_{\max} = \infty$*; equivalently, the charge lattice* $\mathbf{Q}(\mathcal{A}) = 0$ *(no grading kills the quartic pump) and the cubic pump iterates indefinitely.*

The four cases are mutually exclusive and exhaustive under the hypotheses.

Proof. Immediate from the single-line dichotomy plus charge grading. On a primary line, (α, Δ) lies in $\{(0,0), (\alpha, 0), (\alpha, \Delta \neq 0)\}$, giving $r_{\max}|_{\mathrm{prim}} \in \{2, 3, \infty\}$. The three primary-line outcomes correspond to classes G, L, M. Class C is characterised by the *joint* condition: primary line is L-like ($\alpha = 0$ or Jacobi kills S_4) but a nonzero-charge stratum carries $\mathfrak{Q}_{q_\star} \neq 0$. (C3) is the obstruction to the cubic pump: without support of the cyclic complex at charge $2q_\star$, the self-bracket of $\mathfrak{Q}$ vanishes *identically*, not just up to a gauge, and $S_r = 0$ for $r \geq 5$ follows. Non-overlap: G is characterised by $S_3 = 0$ uniformly; L by $S_3 \neq 0$, $S_4|_{\mathrm{prim}} = 0$, no charged stratum; C by (C1)–(C3); M by $S_3 \neq 0$ with no charge-conservation cutoff, forcing $r_{\max} = \infty$. □

Example 6.5 (Examples and witnesses). *Class C witnesses.* $\beta\gamma$ at weight λ carries the $U(1)_{\mathrm{ghost}}$ charge (charge of $\beta = +1$, charge of $\gamma = -1$); the contact invariant lives in charge $q_\star = 0$ via $:\beta\gamma:$ but its quartic pump target has no support at charge ± 2 because $\beta\beta$ and $\gamma\gamma$ OPEs vanish. The bc ghost system is the Koszul dual. Matrix factorisations $\mathrm{MF}(W)$ for quasi-homogeneous W carry the R-charge of the Landau–Ginzburg potential and are class C in the weight-graded sense.

Class C excluding class G: Heisenberg × lattice. $\mathcal{H}_k \otimes V_\Lambda$ at a rank-≥ 1 lattice Λ can carry contact data at nontrivial lattice charge; on the lattice charge-zero sublattice it is Gaussian, so the mixed sector produces genuine class-C behaviour when the OPE of e^α with itself gives a simple pole at $\alpha^2 = 1$.

Theorem 6.6 (Classification of the standard landscape). *The standard chiral algebra families distribute as follows:*

Class	$r_{\max}$	***Families***	***Structural characterization***
G	2	*Heisenberg, lattice VOAs*	*tower terminates at* κ*; higher shadows vanish*
L	3	*affine* $\widehat{\mathfrak{g}}_k$	*Jacobi obstruction is last nonvanishing term*
C	4	$\beta\gamma$*, bc ghosts, matrix factorizations*	*quartic contact class terminates by rank-one rigidity*
M	∞	*Virasoro,* $\mathcal{W}_N$ *(*$N \geq 2$*)*	*all degrees contribute; tower does not terminate*

Proof. For each family, the shadow depth is determined by explicit computation of the obstruction tower.

Class G: The Heisenberg algebra $\mathcal{H}_k$ has a free bosonic OPE $J(z)J(w) \sim k/(z-w)^2$. The shadow metric $Q_{\mathcal{H}}(t) = k^2 + 6kt$ is a perfect square (after completing), so the critical discriminant $\Delta = 0$

and the tower terminates at degree 2: only $\kappa(\mathcal{H}_k) = k$ survives. Lattice VOAs inherit class G from the Heisenberg core.

Class L: For affine $\widehat{\mathfrak{g}}_k$ at generic level, the self-OPE $J^a(z)J^b(w) \sim k\delta^{ab}/(z-w)^2 + f_c^{ab}J^c(w)/(z-w)$ produces a nonvanishing cubic shadow $S_3 \neq 0$ from the structure constants f_c^{ab}. The cubic gauge triviality theorem shows that the cubic MC term is gauge-trivial when $H^1(F^3\mathfrak{g}/F^4\mathfrak{g}, d_2) = 0$, but the associated quartic class lies in a one-dimensional space: the independent sum factorization forces it to vanish, terminating the tower at degree 3.

Class C: The $\beta\gamma$ system has $p_{\max} = 1$ and $k_{\max} = 0$, so no collision residue appears at degree 2. The composite $:\beta\gamma:$ produces a nonvanishing quartic contact class $\mathfrak{Q}_{\beta\gamma}$ at degree 4. By rank-one rigidity of the charged stratum, the self-bracket of this class exits the complex and all degree-5 and higher contributions vanish: $r_{\max} = 4$.

Class M: The Virasoro algebra has $S_3(\mathrm{Vir}_c) = 2$, a nonvanishing c-independent constant (Theorem 6.8 below). The critical discriminant $\Delta_{\mathrm{Vir}} = 40/(5c+22) \neq 0$ for generic c, so the single-line dichotomy forces $r_{\max} = \infty$. The $\mathcal{W}_N$ algebras inherit class M from their Virasoro subalgebra. □

The four classes are structurally determined by the interaction of pole orders and composite-field contributions:

Proposition 6.7 (Depth decomposition). *The total shadow depth decomposes as $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$, where the algebraic depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ takes the values 0 for class G, 1 for class L, 2 for class C, and ∞ for class M. Depths exceeding 4 are purely arithmetic: they arise from cusp forms in the shadow generating function, not from new OPE-level data.*

Proof sketch. Classes G, L, and M are established by the single-line dichotomy and the cubic pump argument above. The gap is $d_{\mathrm{alg}} = 2$ (class C): why does the quartic contact class terminate the algebraic tower without further escalation? For vertex algebras the dichotomy is sharp. Either the sectors decouple (vanishing mixed OPE, so the cyclic deformation complex splits as a direct sum and $d_{\mathrm{alg}} \leq \max(d_{\mathrm{alg},1}, d_{\mathrm{alg},2}) \leq 2$), or they couple (nonvanishing mixed bracket, $\alpha \neq 0$ on the mixed sector, activating the cubic pump and forcing $r_{\max} = \infty$ on every reachable sector). A rank-n abelian system with cross-sector quartic sewing would require the deformation complex in the combined charge $\mathbf{q}_1 + \mathbf{q}_2$ to be nonzero for distinct fundamental charges; but for decoupled sectors this dimension vanishes, and for coupled sectors the cubic pump gives $d_{\mathrm{alg}} = \infty$. Hence only $\{0, 1, 2, \infty\}$ are realized. □

Theorem 6.8 (c-independence of S_3 for Virasoro). *The cubic shadow coefficient of* Vir_c *is*

$$S_3(\mathrm{Vir}_c) = 2, \tag{6.1}$$

independent of the central charge c.

Proof. The cubic shadow coefficient is the ratio $S_3 = (T_{(1)}T)_{\mathrm{lin}}/(T_{(3)}T)_{\mathrm{const}}$ of the linear and constant parts of the depth-2 collision residue. From the Virasoro OPE $T(z)T(w) \sim (c/2)(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1}$, we read $T_{(3)}T = c/2$ and $T_{(1)}T = 2T$, giving

$$S_3 = \frac{2\kappa(\mathrm{Vir}_c)}{\kappa(\mathrm{Vir}_c)} = 2,$$

where $\kappa(\mathrm{Vir}_c) = c/2$ cancels in numerator and denominator. □

Remark 6.9 (Class M membership is finitely witnessed). For the Virasoro algebra, the finite computations $S_3(\mathrm{Vir}_c) = 2$ and $S_4(\mathrm{Vir}_c) = 10/[c(5c+22)]$ at generic c give $\Delta = 8\kappa S_4 \neq 0$. The single-line dichotomy then forces infinite shadow depth. The implication $S_3 \neq 0 \Rightarrow S_4 \neq 0$ is not a general formal rule; here both coefficients are read directly from the Virasoro OPE and the norm of $\Lambda = {:}TT{:} - (3/10)\partial^2 T$.

7 Companion strata: M^* logarithmic and W Wakimoto

The four-class fingerprint (G, L, C, M) stratifies the non-logarithmic non-critical standard landscape by the algebraic invariants (α, Δ) of the shadow metric. Two further strata surface once the landscape is enlarged: logarithmic triplet algebras and screening-kernel free-field realisations.

Definition 7.1 (Class M^*)**.** A chirally Koszul algebra $\mathcal{A}$ belongs to class M^* if (i) the Zhu algebra $A(\mathcal{A})$ is finite-dimensional and (ii) the Hochschild Massey products $\langle \Omega, \Omega, \ldots, \Omega \rangle$ carry amplitude on arbitrarily high tensor powers of the vacuum.

Canonical witnesses are the triplet algebras $\mathcal{W}(p)$ for $p \geq 2$ and the symplectic-fermion pair at $c = -2$. Class M^* shares the mixed-shadow algebraic invariants of class M but is distinguished by Gurarie–Flohr logarithmic amplitudes (Gurarie 1993 arXiv:hep-th/9303160; Flohr 1996 arXiv:hep-th/9605151): Zhu-finite yet Massey-unbounded. The non-logarithmic single-line dichotomy applies to the algebraic invariants, which are unaffected by logarithmic amplitudes; tempering of the shadow tower is blocked.

Definition 7.2 (Class W)**.** A chirally Koszul algebra $\mathcal{A}$ belongs to class W if it is realised as a screening kernel $\mathcal{A} = \bigcap_i \ker Q_{\alpha_i}$ inside a rank-$(N-1)$ Heisenberg algebra, for some finite family of screening operators $\{Q_{\alpha_i}\}$.

Canonical witnesses are the principal $\mathcal{W}_N$-algebras as Feigin–Frenkel–Wakimoto kernels inside $\mathcal{H}^{\oplus(N-1)}$, the $\beta\gamma\phi$ Wakimoto realisation of $\widehat{\mathfrak{sl}}_2$, and admissible affine algebras through parabolic screenings (KRW03, Arakawa07). Class W is not disjoint from G, L, C, M: a class M algebra realised as a Wakimoto kernel inherits Koszulity of the ambient class G parent, and the Drinfeld coproduct descends from the free-field parent to the intersection.

Remark 7.3 (Seven-stratum fingerprint). The stratification is $\{G, L, C, M, \mathrm{FF}, M^*, W\}$. Classes G, L, C, M are determined by the shadow metric on the non-logarithmic non-critical locus. FF is the $\kappa = 0$ boundary. M^* records Massey amplitude behaviour orthogonal to (α, Δ). W records a presentation overlay: a class-L or class-M algebra may also be class-W.

8 The universal shadow-series exponential base

Theorem 8.1 (Universal $C_{\mathcal{A}} = 6$)**.** *Let $\mathcal{A}$ be non-logarithmic class M chirally Koszul with a Virasoro subalgebra at generic central charge. The shadow series $\Sigma_{\mathcal{A}}(z) = \sum_{r\geq 2} S_r(\mathcal{A})\, z^r$ has radius of convergence exactly $1/6$: $C_{\mathcal{A}} := \limsup_r |S_{r+1}/S_r| = 6$. The closed form on the Virasoro primary line is*

$$\Sigma_{\mathrm{Vir}}(z) = \frac{4z^3}{9} - \frac{z^2}{9} + \frac{z}{27} - \frac{\log(1+6z)}{162}.$$

Pole-doubling at all derivatives: $\Sigma^{(k)}_{\mathrm{Vir}}(z) = [P_k + Q_k \log(1+6z)]/(1+6z)^{2k}$.

Remark 8.2 (Decomposition $6 = 3 \cdot 2$). The base 6 factors as the product of the bar-depth seed $r_0 = 3$ (first non-vanishing shadow degree beyond κ) and the Riccati seed coefficient $S_{r_0} = 2$ (equation (6.1)). The same universality extends to $\mathcal{W}_N$ at leading $1/c$; sub-leading corrections $\beta_N = (N+1)(N+2)/2$ asymptote into the universal base.

Proposition 8.3 (Picard–Fuchs for the shadow series). *$\Sigma_{\mathcal{A}}(z)$ satisfies a second-order Picard–Fuchs equation with regular-singular point at $z = -1/6$; the local monodromy is the reflection $z \mapsto -z/(1+6z)$, realising the Borel-plane involution on the shadow tower.*

9 Computational verification

Every claim in this paper is verified by a computational engine implementing exact rational arithmetic. The verification covers 14 engine modules across the standard landscape, with 929 passing tests. The principal verification targets are:

Family	$(p_{\max}, k_{\max}, r_{\max})$	**Class**	**Tests**
Heisenberg $\mathcal{H}_k$	$(2, 1, 2)$	G	67
Lattice VOA V_Λ	$(2, 1, 2)$	G	41
$\widehat{\mathfrak{sl}}_2$	$(2, 1, 3)$	L	89
$\widehat{\mathfrak{sl}}_3$	$(2, 1, 3)$	L	78
$\widehat{\mathfrak{sl}}_N$, $N = 4, \ldots, 8$	$(2, 1, 3)$	L	112
$\beta\gamma$	$(1, 0, 4)$	C	53
Vir_c	$(4, 3, \infty)$	M	154
$\mathcal{W}_3$	$(6, 5, \infty)$	M	131
$\mathcal{W}_N$, $N = 4, 5, 6$	$(2N, 2N{-}1, \infty)$	M	204

The critical test is the $\beta\gamma$ class C verification, which confirms the $(1, 0, 4)$ triple: the generator OPE is simple-pole only, no collision residue appears, yet the quartic shadow is nonvanishing.

Cross-family consistency checks enforce the structural constraints: the relation $k_{\max} = p_{\max} - 1$ is verified for every family; the shadow class assignment is cross-checked against the critical discriminant Δ; and the depth decomposition $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ is verified numerically at multiple parameter values for each family.

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